CoM³eT: A foundation model for medical image analysis through federated, multidimensional context integration

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Abstract

Medical foundation models improve generalization when training AI models with limited labeled data, but remain confined to a single specialty, such as pathology or radiology, and to either sparse or dense outputs, such as classification or segmentation. Here, we present CoM³eT (**Co**-representation **M**ultidimensional **M**ultitask **Me**dical **T**ransformer), a medical vision foundation model that unifies pathology and radiology, sparse and dense predictions, and two- and higher-dimensional inputs by modeling multidimensional context with attention. CoM³eT outperformed other medical foundation models in an open competition spanning five tomographic, four whole-specimen, and three two-dimensional datasets, covering sparse and dense prediction tasks as well as report generation. When adapted across diverse clinical applications, training fewer than 2.5% of parameters achieved performance comparable to full fine-tuning, enabling research without access to high-performance GPU clusters. Applied to federated learning across hospitals, this approach achieved performance comparable to pooled-data training over internet connections and with consumer-grade hardware.

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Main

Artificial intelligence (AI) is improving the analysis of medical data, enabling more accurate diagnoses and more personalized therapy decisions. In particular, deep learning can turn massive datasets into multi-purpose models. However, high-quality, large-scale imaging datasets are unavailable for many diseases, which limits model generalization.

To address data scarcity, vision foundation models (FMs) can reduce task-specific data requirements by pretraining on diverse datasets. For example, VIRCHOW[1] is a vision transformer trained on 1.5 million histopathology images from a single modality and stain, and CT-FM[2] performs strongly across multiple CT tasks after pretraining on 148,000 CTs. Such medical FMs are typically limited to single specialties such as pathology or radiology, or even single modalities, and thus cannot jointly analyze a patient's multimodal images[3]. The limited scope of medical FMs could stem from the difficulty of curating general pretraining datasets. Datasets from textbooks (e.g., PathCAP[4]) or social media (e.g., PLIP[5]) may not align with clinical practice due to selection biases. Alternatively, datasets mined directly from clinical information systems achieve large quantities and full image quality, but often originate from one or a few sites[1,2]. Because data must be pooled for pretraining, these datasets tend to be narrow and fail to capture the diversity of global populations.

Federated learning lets institutions train models collaboratively without pooling data. This can represent population diversity better and reduce the need for central data-pooling infrastructure. However, large models make this difficult. For example, federated averaging[6] requires repeated exchange of the full model between participating sites and a central server, which is impractical for large models. Parameter-efficient fine-tuning addresses this by reducing the number of trainable parameters while achieving near-centralized performance[7]. However, if the fine-tuned parameters are distributed throughout the model, the full computation must run at every site, requiring powerful hardware everywhere. Instead, partial fine-tuning updates only top layers while keeping the base of the model frozen, enabling most of the computation to be performed only once.

Besides being restricted in data origin, existing models are typically applicable to either sparse tasks such as classification and regression, or dense tasks such as segmentation and instance detection, whereas general medical vision FMs require both. UNI[8] is a vision transformer for computational pathology that extends to segmentation by adding dense adapters. However, due to the non-hierarchical nature of transformers, segmentation performance lags behind specialized architectures, requiring a relatively complex dense adapter and a large number of segmentation-specific parameters (up to 50%), which makes them less data-efficient[9]. In segmentation, nnU-Net[10] has demonstrated the continued strength of the traditional hierarchical U-Net but it uses a manually defined pipeline with slightly different architectures for different tasks and is therefore not universally pretrainable.

Beyond being limited to either sparse or dense tasks, handling many data types challenges medical vision FMs. Medical images come in diverse formats, including high-resolution 2D images (e.g. X-rays and tissue microarrays), gigapixel images (e.g. whole-slide images in pathology), and volumetric images (e.g. CT and MRI scans). A solution is to decompose them into their greatest common denominator: 2D image patches, such as frames in videos, slices in tomographic images, and tiles in gigapixel images. For example, SLIViT[11] approached 3D classification using a 2D encoder for feature extraction and a transformer for aggregation. Similarly, CAT-Net[12] introduced a cross-slice transformer for 3D segmentation that, unlike 3D networks, maintained its performance on flat/anisotropic volumes while enabling long-range information exchange between slices.

In this article, we show that a universal medical vision FM for multidimensional images (CoM³eT; **Co**-representation **M**ultidimensional **M**ultitask **Me**dical **T**ransformer) can be trained by combining a strong image encoder with transformer modules. While CAT-Net is limited to segmentation and SLIViT to classification, CoM³eT processes both for multidimensional images using co-representations of semantic and spatial information, referred to as patch and pixel tokens.

First, we constructed CoM³eT by scaling multitask pretraining[13] to over 60 tasks. We introduced attention-based components that handled both sparse tasks (e.g., classification) and dense tasks (e.g., segmentation) for multidimensional images in one model while sharing most parameters across tasks. Finally, we developed dimensionality-agnostic positional encoding for multitask FMs. For each contribution, we constructed a synthetic benchmark to isolate its effect and quantified its impact on real multidimensional images with a representative test task. The study is summarized in Fig. 1. To compare with state-of-the-art FMs, we submitted CoM³eT to the UNICORN competition for medical FMs, where it ranked first overall and first in both radiology and pathology[14]. Then, we showed that partial fine-tuning was effective even for applications dissimilar to the pretraining tasks. By reducing training resources and communication load, partial fine-tuning enabled federated fine-tuning, which we demonstrated in a real-world setup with German university hospitals.

Results

CoM³eT unifies multidimensional medical imaging tasks in a single model

CoM³eT reuses shared modules across tasks. The largest is the vision backbone, shared by all tasks. It processes image inputs and produces patch tokens, which can represent full 2D medical images such as X-rays, frames in videos, slices in tomographic images, or patches in gigapixel images. Patch tokens are CoM³eT's fundamental units, and groups of patch tokens represent multidimensional inputs. For segmentation tasks, semantic information in the patch tokens alone is not sufficient because segmentation also requires spatial information. Therefore, the vision backbone also produces feature pyramids, i.e., multi-scale feature maps that preserve spatial layout across resolutions, as is common in segmentation architectures.

The image transformer models interactions among sets of patch tokens while preserving one contextualized patch token per input image patch. This lets the model solve both patch-level tasks, such as identifying slices that contain tumor, and patient-level tasks, such as cancer grading. The pyramid transformer extends the image transformer to dense tasks and uses its output tokens to enrich the vision backbone's feature maps with global context. Subsequently, the decoder converts these into fine-grained hyperpixel tokens. Thanks to this modularity, CoM³eT can support both patch-based and multidimensional imaging data, as well as sparse and dense prediction tasks. Table 1 lists sizes of the model variants, Fig. 2 provides an overview of the architecture, and Extended Data Fig. 1 illustrates how CoM³eT was verified to use local and global context and positional information for classification and segmentation on synthetic data.

CoM³eT leads the UNICORN competition

Meaningful comparisons between medical FMs require an independent benchmark based entirely on private data[15], evaluated without internet access to avoid data leakage. Such a benchmark should cover the main medical imaging task types, including classification, pixel-level prediction such as segmentation, and vision-language reporting, as well as the main imaging domains: pathology with gigapixel images or large 2D tiles, and radiology with 3D images such as CT and multiparametric MRI. UNICORN[14] is the first benchmark fulfilling these criteria. It uses a frozen setting: model weights remain fixed and only lightweight adapters are trained on extracted representations.

Across the benchmark, CoM³eT was the only model applicable to all vision and vision-language tasks, achieving the best performance on 6 of 12 tasks. In addition, CoM³eT (0.442 ± 0.022) outperformed a theoretical best ensemble using the strongest competing FM for each task (0.357 ± 0.014; Fig. 3 A, left).

CoM³eT also achieved the best average performance within each domain. In pathology (0.482 ± 0.018; Fig. 3 A, middle), it was best in 4 of the 6 tasks, including biochemical recurrence prediction in prostatectomy patients, tumor proportion score in IHC-stained lung cancer samples, mitotic figure detection in breast cancer H&E-stained WSIs, and tissue type segmentation. The runner-up methods were an ensemble of TITAN[16] and CONCH[17] (0.356 ± 0.015), followed by UNI2[8] (0.348 ± 0.014). In radiology (0.458 ± 0.027; Fig. 3 A, right), the next best method[18] reached 0.341 ± 0.012 with an ensemble of two adapted nnU-Net models[10] pretrained on CT and MRI. Notably, only CoM³eT solved the multiparametric MRI segmentation task above random-guessing level, consistent with its ability to integrate multiple inputs at the same spatial position. An ensemble of models restricted to 3D patches rather than voxel-level features performed substantially worse (0.077 ± 0.006)[2,19,20].

Multidimensional imaging increases supervised pretraining data volume

CoM³eT's first pretraining stage used natural-image datasets to learn general representations. ImageNet[21,22] contributed more than 12 million classification images, and COCO[23] added more than 200,000 images for vision-language alignment, object annotations, and object supercategories. The primary stage then added medical data from pathology and radiology. Pretraining for pathology covered more than 10,000 whole-slide images and more than 100,000 annotated patches. In radiology, data ranged from small segmentation cohorts of a few dozen to a few hundred 3D scans up to large public resources with thousands of examinations, while meta-learning datasets such as RadImageNet alone contributed more than 1.3 million labeled images. Taken together, this stage combined supervised data from more than 100,000 patients across collections spanning single-cell images, histology patches, whole-slide images, radiographs, CT, endoscopic images, ultrasound, and MRI.

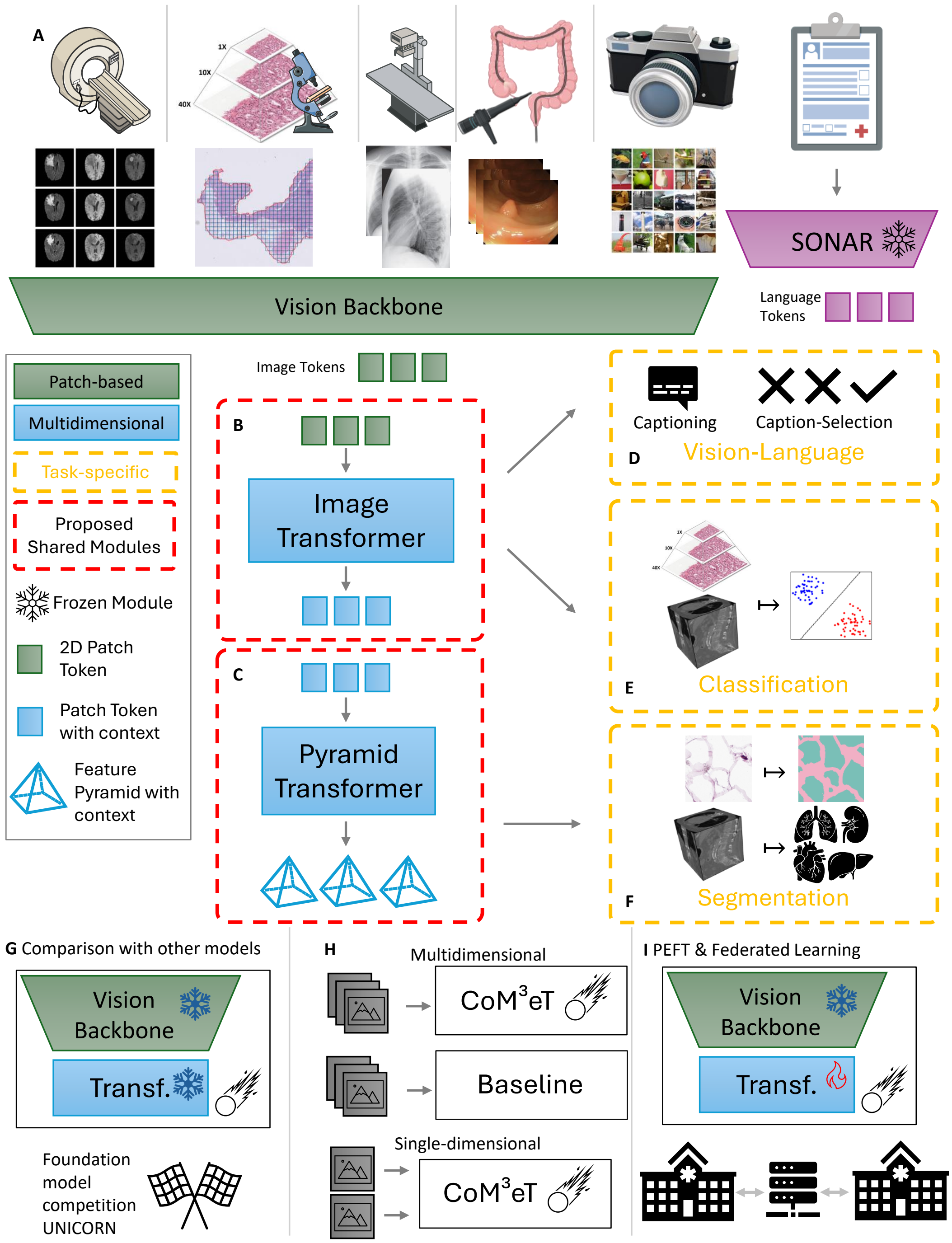

Figure 1: Study overview. **A,** CoM³eT's pretraining includes multidimensional images from multiple medical imaging modalities together with natural image data into a single model via multitask learning. **B,** the image transformer models global context from outputs of the vision backbone. **C,** the pyramid transformer uses this context to extend FM segmentation to multidimensional images such as 3D volumes. **D-F,** pretraining uses supervised labels across vision-language, classification and segmentation tasks. **G,** CoM³eT participated in UNICORN, the first benchmark competition for medical FMs, which compared frozen FMs on a diverse set of medical imaging tasks. **H,** performance comparison between CoM³eT, a specialized reference model, and CoM³eT restricted to individual patches. **I,** CoM³eT's architecture supports partial fine-tuning. We compare full and partial fine-tuning and apply partial fine-tuning in a real-world federated learning setting.

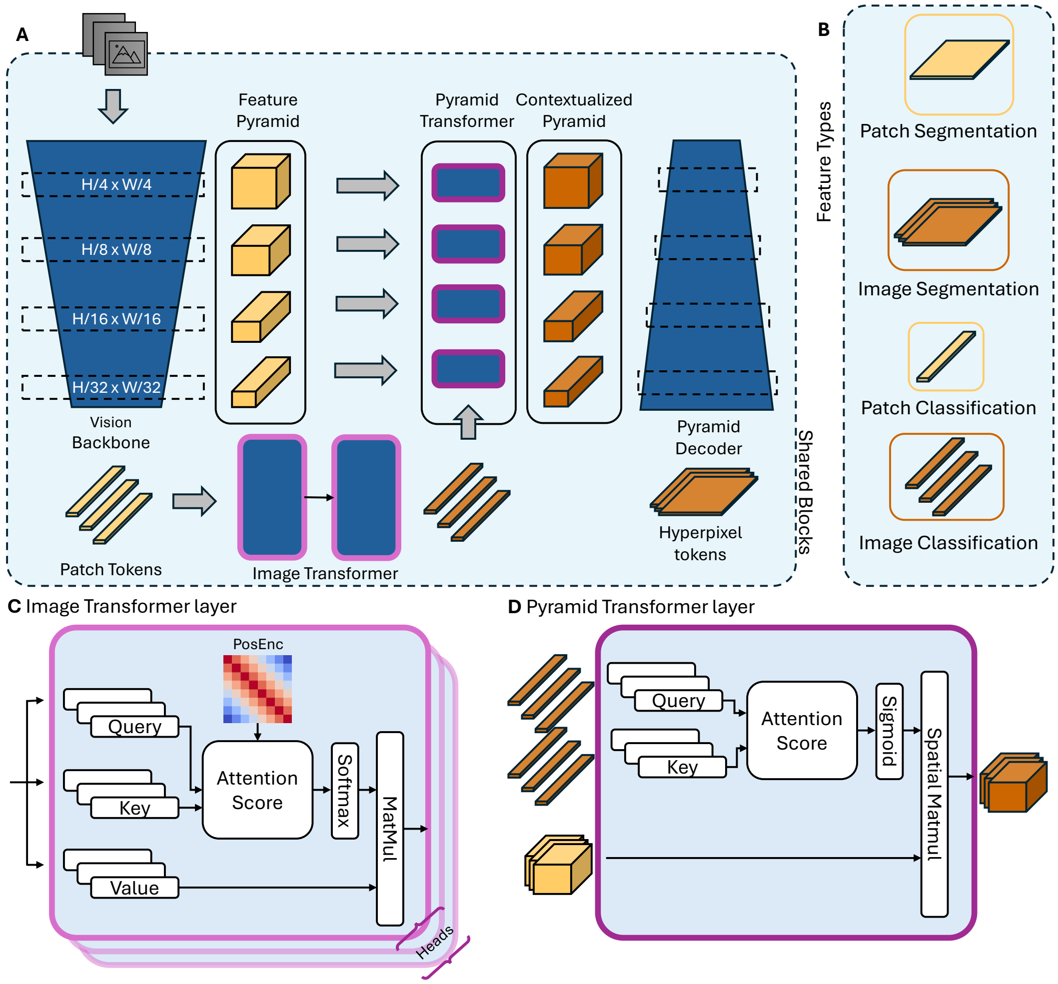

Figure 2: Model overview. A, First, CoM³eT's vision backbone encodes image inputs, representing a multidimensional medical image as a group of patch tokens and feature pyramids. In the case of multidimensional images, the image transformer then computes global context. For multidimensional segmentation tasks, the pyramid transformer uses the image transformer's output tokens to enrich the feature pyramids with global context. **B,** CoM³eT creates two types of features. Segmentation tasks use features with their spatial shape intact (hyperpixel tokens), while classification tasks use feature vectors (patch tokens). **C,** The image transformer is an encoder-only transformer that models interactions between patch tokens to add global multidimensional context while preserving one representation per input. **D,** The pyramid transformer uses the outputs of the image transformer to inject global multidimensional context into feature maps.

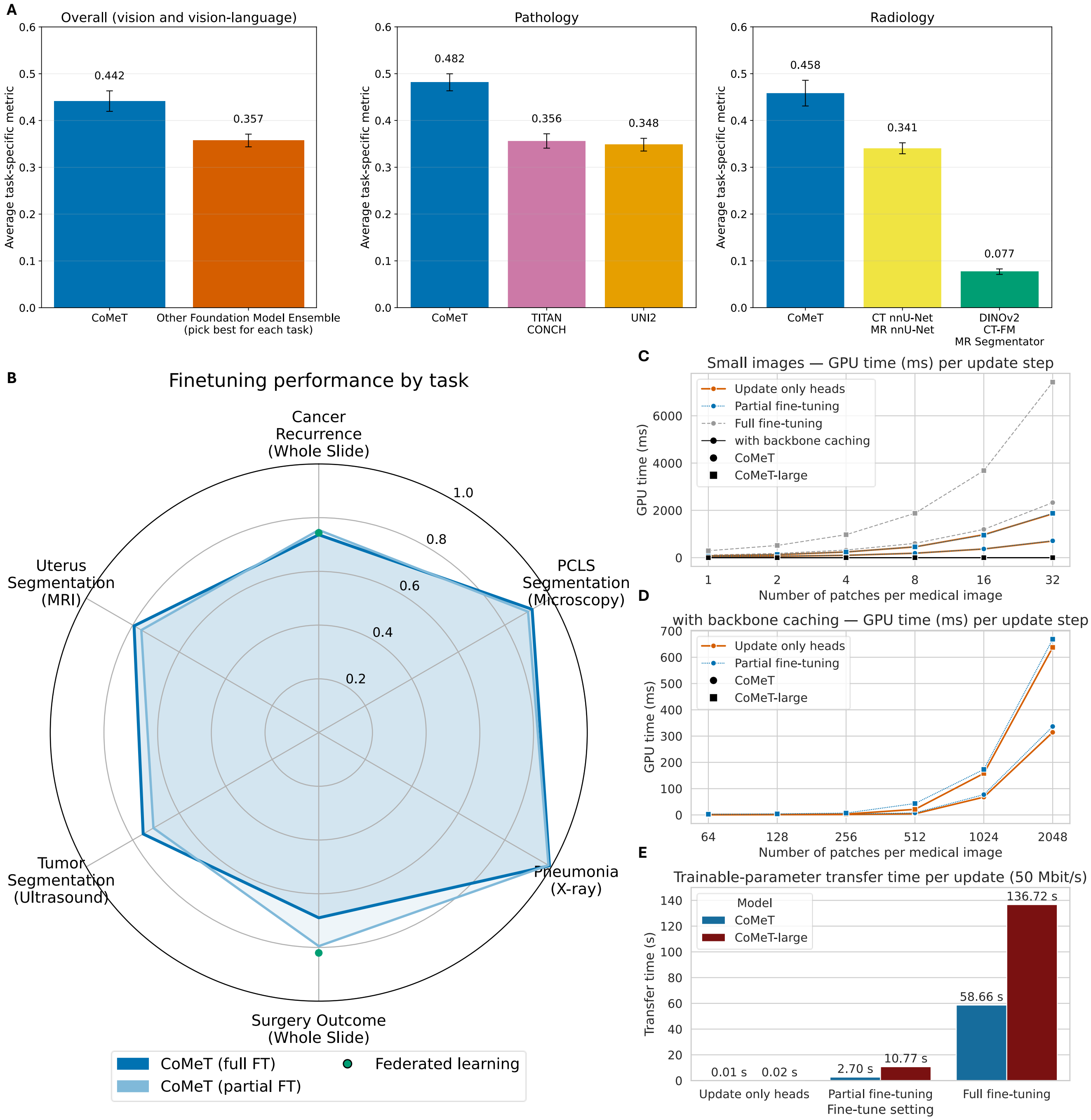

Figure 3: Results overview. A, Mean task-specific metrics from the UNICORN competition (each task's metric scaled to the range from random-guessing performance to maximum performance). Left, CoM³eT compared with a theoretical best ensemble composed of the strongest competing FM for each task. Middle and right, pathology and radiology results, respectively, compared with the second- and third-place FMs or combinations of FMs. **B,** Fine-tuning results with partial and full fine-tuning. **C,** GPU time, measured on an A100, for different fine-tuning settings. For multidimensional images with between 1 and 32 patch tokens, training based on cached patch tokens (backbone caching) has almost zero cost for partial fine-tuning (black). The blue and orange lines overlap because partial fine-tuning and no fine-tuning incur nearly identical cost. **D,** GPU time for multidimensional images with 64 or more patch tokens. Full fine-tuning cannot be used for these images. **E,** Time required per update for each fine-tuning setting for CoM³eT and its larger variant, CoM³eT-Large. In federated learning, a full update step requires exchanging the full model state twice, first from the sites to the server and then from the server back to the sites. Unless stated otherwise, "CoM³eT" refers to the CoM³eT-Base variant.

We quantified the image transformer's performance for new medical applications via fine-tuning, comparing our multidimensional approach with averaged 2D processing and, where applicable, a specialized single-task method.

Pathology tasks quantified the image transformer's effect where each whole-slide image was represented by many patches. "Cancer Recurrence (Whole Slide)"[24] predicted time to biochemical recurrence after prostatectomy. CoM³eT achieved an agreement between predicted and observed event times (C-index) of 0.736 ± 0.020 with the image transformer and 0.644 ± 0.040 without it. "Surgery Outcome (Whole Slide)"[25] predicted whether biochemical recurrence would occur within one year after prostatectomy. Here, performance improved from an AUC of 0.607 ± 0.176 without the image transformer to 0.690 ± 0.152 with it. Together, these results show that the image transformer has a strong positive effect on pathology tasks.

FMs often perform best in data-scarce settings. Here, however, we tested whether CoM³eT's unified architecture remained competitive with specialized models on a large task-specific dataset spanning 1,330 imaging studies from more than 10 hospitals[26]. We evaluated it on "Tumor Classification (MRI)", distinguishing benign from malignant breast lesions. Among the tested specialized models, a video Swin Transformer[27] performed best, with an AUC of 0.840 (95% CI: 0.796 to 0.879). CoM³eT with its image transformer achieved an AUC of 0.859 (95% CI: 0.818 to 0.897). Without the image transformer, performance dropped to 0.782 (95% CI: 0.732 to 0.829).

The pyramid transformer extends CoM³eT to multidimensional segmentation tasks

When analyzing volumetric images, clinicians rely on context beyond individual slices. Both nearby slices and global context can indicate the presence of a structure. For CoM³eT, the pyramid transformer provided the attention mechanism that combines image patches for multidimensional segmentation while reusing most parameters with other multidimensional tasks.

First, synthetic experiments verified that local and global context and positional information could be used for segmentation Extended Data Fig. 1. Then, we fine-tuned CoM³eT on three 3D segmentation settings that differed in dataset size and context requirements. Representative inputs and task groupings are shown in Fig. 4.

On "Uterus Segmentation (MRI)", a segmentation task for myometrium, junctional zone, and endometrium, CoM³eT already exceeded the strongest specialized baseline without the pyramid transformer ($75.49\% \pm 0.5\%$ 3D Dice versus 71.0% for 3D U-Net[28]) on this anisotropic dataset with 13 patients. Adding the pyramid transformer further improved performance to $79.4\% \pm 0.3\%$.

For "Vessel Segmentation (CT)", a segmentation task for the aorta and its branches that was expected to depend strongly on local continuity across neighboring slices, CoM³eT performed on par with nnU-Net[10] even without the pyramid transformer (80.45%, 95% CI: 79.02% to 81.47% 3D Dice versus 80.38%, 95% CI: 78.33% to 81.36%). With the pyramid transformer, performance increased to 82.11% (95% CI: 80.70% to 83.17%), indicating that added multidimensional context improves even tasks that are challenging for a model that aggregates information at the patch token level.

The "Tumor Segmentation (MRI)" task was to segment breast lesions in contrast-enhanced MRI using a multicenter dataset of 3,936 patients. This tested whether CoM³eT's unified architecture remained competitive outside the data-scarce regime in which FMs often perform best. Overall, nnU-Net[10] achieved a 33.13% 3D Dice score (95% CI: 30.71% to 35.51%), whereas CoM³eT reached 25.28% (95% CI: 22.86% to 27.73%) without the pyramid transformer and 58.62% (95% CI: 55.92% to 61.31%) with it. This large gain suggests that CoM³eT has an advantage when it must both identify and segment lesions. When we simplified the task to images with lesions only, Dice scores became similar between methods, at 53.0% for nnU-Net (95% CI: 50.5% to 55.4%) and 53.3% for CoM³eT (95% CI: 50.7% to 55.9%). Tumor size has traditionally been summarized by the largest diameter, but automatic segmentation makes three-dimensional volume practical. The average volume error was 4.23 ml (95% CI: 3.57 to 4.96) for nnU-Net and 4.46 ml (95% CI: 3.73 to 5.28) for CoM³eT without the pyramid transformer, which the pyramid transformer reduced to 2.15 ml (95% CI: 1.65 to 2.75).

Partial fine-tuning of CoM³eT is computationally efficient across various tasks and domains

We compared full and partial fine-tuning on the same tasks and hardware. For partial fine-tuning, we froze the vision backbone and trained the other components, including the image transformer, pyramid transformer, and decoder. This reduced CoM³eT's trainable model size by 97.5%. It also lowered the time required for an update step at 32 patch tokens per image from 2330ms to 712ms. Freezing the vision backbone also allowed caching of

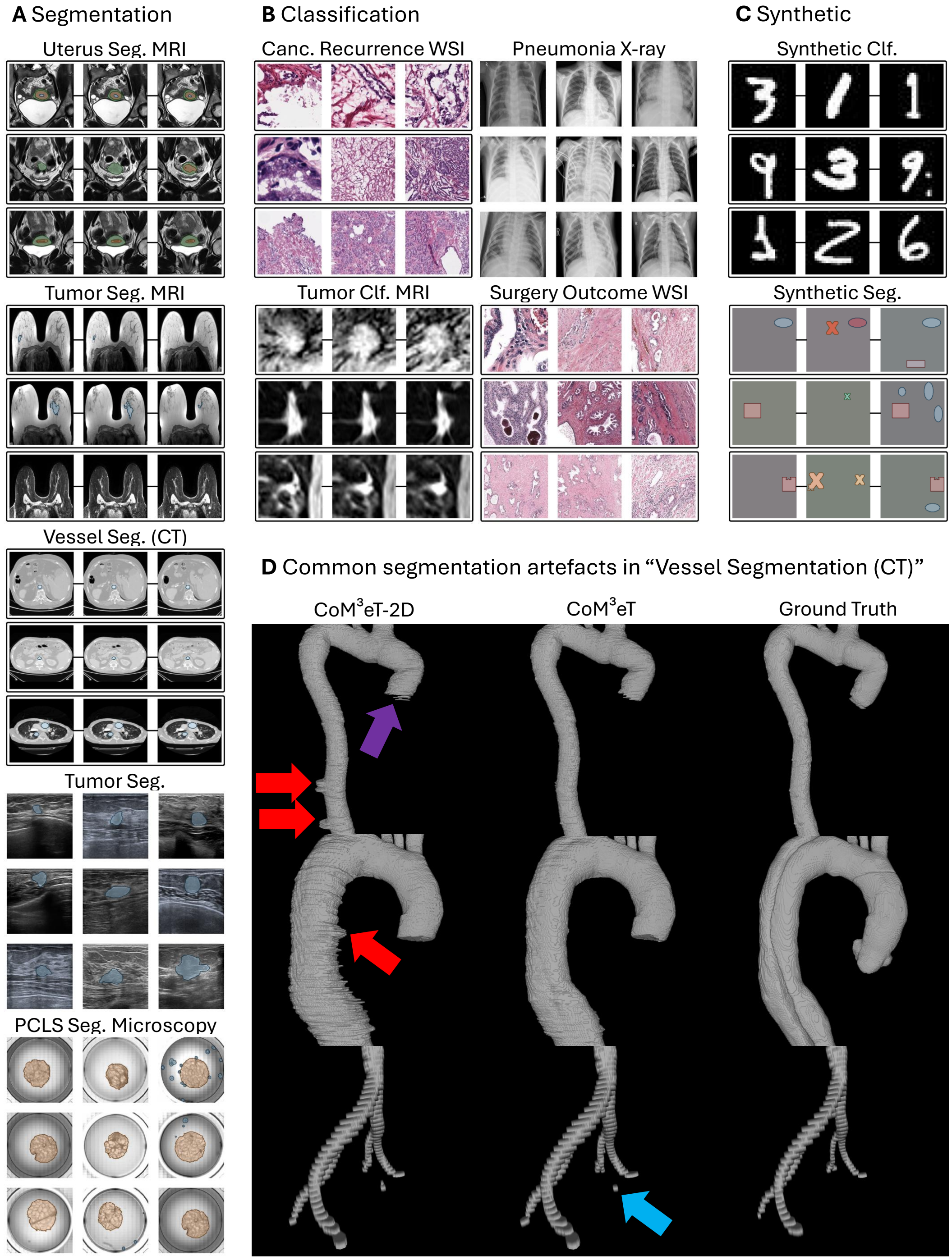

Figure 4: Test data. A, The study included three 3D and two 2D segmentation tasks. Grouped inputs are shown as rows within boxes, and tasks that depend on positional information are connected by a line. **B,** Four classification tasks were included, one of them ungrouped. **C,** Synthetic tasks supported model development and ablation studies. **D, Segmentation results.** Without the image and pyramid transformer (CoM^{3}eT-2D), common problems include discontinuities in tubular structures (red) and missing segments in uncertain regions (purple), other errors remain for CoM^{3}eT (blue).

patch tokens, which reduced this time by $> 99.9\%$ compared with full fine-tuning (from 2.3 seconds to 1.5ms, see Fig. 3 C-E).

Across all tasks, an equivalence test supported equivalence between full and partial fine-tuning ($p_{\text{lower}} < 0.001$, $p_{\text{upper}} = 0.008$). Neither strategy failed on any of the tested datasets. Mean performance was similar in both settings, at 0.833 for full fine-tuning and 0.839 for partial fine-tuning.

Whole-slide image analysis is a relevant test case for partial fine-tuning because caching of patch tokens offers large practical benefits in this setting. For "Cancer Recurrence (Whole Slide)", CoM³eT reached a C-index of 0.736 ± 0.020 with full fine-tuning and 0.754 ± 0.012 with partial fine-tuning. On "Surgery Outcome (Whole Slide)", CoM³eT reached an AUC of 0.690 ± 0.152 with full fine-tuning and 0.795 ± 0.170 with partial fine-tuning. Both tasks showed a trend toward better performance with partial fine-tuning. For the two whole-slide tasks, we also compared attention topology augmentation with full attention in the image transformer during partial fine-tuning. With full attention, CoM³eT reached a C-index of 0.757 ± 0.015 on "Cancer Recurrence (Whole Slide)" and an AUC of 0.771 ± 0.160 on "Surgery Outcome (Whole Slide)". A one-sided paired t-test did not show that attention topology augmentation outperformed full attention ($p = 0.351$).

Volumetric segmentation was included to quantify partial fine-tuning for 3D segmentation, which relies on both the image transformer and the pyramid transformer. For the "Uterus Segmentation (MRI)" volumetric segmentation task involving three structures, the myometrium, the junctional zone, and the endometrium, CoM³eT achieved a mean Dice score of $79.4\% \pm 0.3\%$ with full fine-tuning and $76.3\% \pm 0.4\%$ with partial fine-tuning, both exceeding the 3D U-Net baseline of 71.0%[28].

We also evaluated partial fine-tuning on patch-level 2D tasks. Here, the image transformer and decoder were partially fine-tuned. Because the group size was one, the image transformer was equivalent to a pretrained MLP. We evaluated three tasks: "Pneumonia (X-ray)", a chest X-ray classification task, "PCLS Segmentation (Microscopy)", a brightfield microscopy tissue and artifact segmentation task, and "Tumor Segmentation (Ultrasound)", an ultrasound lesion segmentation task. In "Pneumonia (X-ray)", which required only weak generalization because chest X-ray data were included in pretraining, CoM³eT reached an AUC of 0.992 ± 0.002 with full fine-tuning and 0.988 ± 0.003 with partial fine-tuning. On "Tumor Segmentation (Ultrasound)", which tested generalization because no ultrasound segmentation data were included in pretraining, CoM³eT reached a Dice score of $75.5\% \pm 2.6\%$ with full fine-tuning and $71.1\% \pm 2.3\%$ with partial fine-tuning. The "PCLS Segmentation (Microscopy)" task involved segmentation of tissue and artifacts to enable monitoring of tissue characteristics over time. Despite including no related data during pretraining, CoM³eT achieved a mean Dice score of $91.8\% \pm 4.7\%$ with full fine-tuning and $90.0\% \pm 4.5\%$ with partial fine-tuning.

Successful federated fine-tuning of CoM³eT in real multi-site scenarios with consumer-grade hardware and over internet connections

Federated learning followed our partial fine-tuning setup, except that synchronization occurred every 4 local training steps and each site updated the model only on its local data subset. This reduced local compute and synchronization costs (see Fig. 3 C-E). Keeping the vision backbone frozen allowed caching and reusing patch tokens during local training, while synchronizing only the trainable image transformer parameters across sites. We deployed federated fine-tuning in a three-site study with two German university hospitals (UKFFM and Charité) and one site contributing public biochemical recurrence data. The system ran on consumer-grade GPUs over the internet. Federated fine-tuning started from an earlier version of CoM³eT which did not include pretraining tasks that used biochemical recurrence labels. For the centralized baseline (CoM³eT) all data were pooled.

Model selection was based on the averaged local validation results at each site (UKFFM: 0.754, Charité: 0.738, Public: 0.727), resulting in CoM³eT-FL. CoM³eT-FL achieved a C-index of 0.743 ± 0.012 on "Cancer Recurrence (Whole Slide)"[24] and an AUC of 0.820 ± 0.125 on "Surgery Outcome (Whole Slide)"[25]. For the corresponding pooled-data CoM³eT achieved a C-index of 0.754 ± 0.012 and an AUC of 0.795 ± 0.170, respectively. An equivalence test supported equivalence between CoM³eT-FL and CoM³eT, ruling out a difference of 0.05 or more in either direction ($p_{\text{lower}} = 0.019$, $p_{\text{upper}} = 0.032$), suggesting that federated partial fine-tuning can achieve similar performance to centralized training in these applications.

Discussion

Traditionally, medical vision has required different model families for different image and task types. For example, nnU-Net[10] has become a general framework for many medical imaging problems, but its training is restricted to a single segmentation dataset. Cerberus[29] supports both classification and segmentation, but only for 2D histology patches. CT-FM[2] incorporates 148,000 scans, but is specific to 3D CT, whereas the million-image-scale model VIRCHOW[1] is specific to histology images. CoM³eT, however, has demonstrated that a single architecture can handle these diverse image and task types, enabling simultaneous pretraining on standard 2D images and multidimensional data. This flexibility enabled CoM³eT to scale to additional pretraining tasks, ultimately winning the UNICORN competition[14] for medical FMs, where it was also the only model that, on its own, was applicable to all vision and vision-language tasks. However, additional data types and domains compatible with CoM³eT, such as video-based tasks and optical coherence tomography, are not yet used.

Previously, foundation models were often pretrained with a single type of supervision, such as self-supervised learning[2,8] or text supervision[30]. Other approaches use separate stages to combine multiple levels of supervision. For example, pathology-specific foundation models[16,31] commonly require training a patch encoder as a first stage. However, sequential training requires the first stage to encode all information useful for later stages as well. TITAN[16] then adds more self-supervised and language-supervised stages, which could also cause catastrophic forgetting because the same modules are trained sequentially. For example, fine-tuning the general-purpose model Gemini[32] for medical tasks degraded performance on general tasks, including medically relevant ones such as management plan appropriateness[33]. Rather than integrating training objectives sequentially onto a self-supervised stage, CoM³eT achieves high-performing representations through a single final pretraining stage that spans all available data and all levels of supervision, using structured labels where available and text supervision otherwise. This challenges the weight the field places on an initial, or even single, self-supervised pretraining stage. Still, expanding with more weakly and self-supervised pretraining strategies is a promising future direction.

Segmentation objectives give a clear training signal for CoM³eT's hyperpixel tokens. These pixel-level representations maintain spatial context beyond 2D and 3D images by application of the proposed pyramid transformer, generalizing what task-specific models such as CAT-Net[12] achieved with 2D pretrained modules for anisotropic 3D segmentation. For example, the UNICORN competition[14] showed that CoM³eT is the only multi-purpose model applicable to multiparametric MRI (4D) because it can process multiple 3D volumes for segmentation. In our own evaluation, we used vessel segmentation in CT to judge the pyramid transformer's ability to follow local continuity across neighboring slices. Qualitatively, segmentation of multiple small vessels on the same slice remains suboptimal, possibly because the pyramid transformer restricts attention to slices rather than individual voxels. However, since this work focused on evaluating CoM³eT's feature quality, we did not optimize the task heads. For example, using 3D convolutional heads could already improve performance.

Real-world federated partial fine-tuning of CoM³eT achieved performance comparable to centralized training, addressing key problems in developing medical AI: limited access to labeled data, significant computational requirements, and privacy concerns[34]. However, cybersecurity, connection timeouts, and data access policies posed significant challenges that models cannot address. Without standardization and trusted training infrastructure, federated learning will continue to require manual engineering effort and supervision.

A promising next step is to use CoM³eT as the vision component of conversational assistants. By aligning CoM³eT's embedding space with a language model, this could address shortcomings in visual performance that currently limit vision-language models[33,35,36], supplying the specialized representations that general models lack. CoM³eT is equally suited to agentic systems, which rely on external tools to act on their environment: its auxiliary task heads form a reusable library of image analysis tools that such systems could combine into a cohesive care plan[15]. Because partial and federated fine-tuning are fast and resource-efficient, this library could be extended with new vision tools trained on multi-centric data from clinical information systems, including tasks not covered by the current pretraining set.

Methods

Model architecture

We used multitask pretraining[13,37], in which the model was structured into shared modules (components used across tasks) and task heads (components not used across tasks) reflecting increasing levels of supervision, ranging from language to structured tasks to pixel-level tasks. Each task consisted of a training objective, a dataset, a target definition, and a linear task head. A task's target definition specified what the model predicted (for example a classification label, a segmentation mask, or a detection target) and how the loss was computed. The pretraining alternated between all tasks, forcing the shared modules to learn generally applicable features.

CoM³eT's architecture differed in several ways. For CoM³eT, we adapted the size of the vision backbone to test its scalability and increased the number of parameters as shown in Table 1. We also added the image transformer, a transformer module that establishes multidimensional context from the encoded patch tokens, and the pyramid transformer module for dense multidimensional segmentation. Together, these modules allowed CoM³eT to model relationships across image patches and propagate this context back into dense spatial predictions. Using a 2D vision backbone as a central component also enables to simplify the architecture for true 2D problems when needed.

Following this design, the vision encoder accepted 2D RGB images with shape $(3, H, W)$, where H and W denote variable spatial dimensions. The method is compatible with any vision encoder that supports this input format and produces hierarchical feature maps, such as $[(C, \frac{H}{2}, \frac{W}{2}), (2C, \frac{H}{4}, \frac{W}{4}), ...]$. The number of channels C (e.g., 64, 128, ...) depends on the encoder variant. Here, we selected Swin Transformer V2[38] as vision backbone, as these models scale effectively to large parameter counts and are not restricted to a fixed image size. To obtain a compact representation for each patch token, we applied global average pooling followed by a linear projection for dimensionality reduction. This reduction was especially helpful when processing medical images composed of many patch tokens. Details on parameter counts and dimensionality are summarized in Table 1.

Table 1: Architecture statistics. Each 2D input (such as an image or image patch) is described by the "Patch token size", which refers to the dimension of the feature vector (patch token) produced by the encoder for that input. "Hyperpixel token size" refers to the dimension of the hyperpixel token (the embedding corresponding to each spatial location within the input) as produced by the decoder.

CoM³eT	Vision backbone		#Parameters transformer		Dimensionality	
Variant	Variant	#Parameters	Image T.	Pyramid T.	Patch token	Hyperpixel token
Tiny	Swin-T	27,582,570	1,054,208	529,120	256	32
UNICORN	Swin-B	86,905,848	4,205,568	2,105,472	512	32
FL	Swin-B	86,905,848	4,205,568	2,105,472	512	64
Base	Swin-B	86,905,848	4,205,568	2,105,472	512	64
Large	Swin-L	195,220,980	16,799,744	8,403,904	1024	128

The image transformer was introduced for modeling the interactions and establishing context between different patch tokens (the fundamental unit for our training method such as the representation of a 2D image patch). It generated representations for all patch tokens, which generalized previous aggregation methods[11] to multidimensional medical images for simultaneous processing of dense prediction tasks such as 3D segmentation and sparse tasks such as classification. For its architecture, we adopted the general deep bidirectional encoder of BERT[39], which was originally designed as a pure language model operating over language tokens. Here, instead of language tokens, we utilized patch tokens generated by the vision backbone, extending BERT's architecture to process visual information. Unlike BERT, our pretraining did not require masking input tokens because each patch token could attend globally. However, multidimensional medical datasets often contain few cases with many patch tokens each, which increases the risk of memorizing individual patients or shortcut features such as pen markings in gigapixel images. To regularize this setting, we introduced attention topology augmentation: for multidimensional images, we randomly dropped connections in the image transformer so that each output patch token attended only to a random subgroup, reusing the same mask across attention heads and applying a rate of 0.5 for all multidimensional classification tasks during training and fine-tuning.

Since transformers are permutation-invariant, positional encoding is required when position is not directly visible in the image. For volumetric images such as CT and MRI, we chose relative rather than absolute encoding under the hypothesis that modeling pairwise slice relationships is sufficient to capture relevant spatial context. For CoM³eT, we chose ALiBi[40], which handles variable sequence lengths and both short- and long-range dependencies. Because ALiBi is originally asymmetric, we implemented a symmetric variant so that the attention bias is the same from position A to B and from B to A, and extended it to support non-consecutive and duplicate positions (e.g. for multiparametric MRI). To remain robust across the different scales of medical imaging, we encoded distance not in physical units but as the number of patch tokens between two positions. Thus, adjacent patch tokens had a distance of 1, while two patch tokens with three tokens in between had a distance of 4.

To tune the method and test correctness, we created a synthetic task from MNIST (28×28 grayscale digit images). For training, we sampled ordered groups of eight random digits (see Extended Data Fig. 1 A) and added relative symmetric ALiBi positional encoding between patch tokens. We designed this task to test both local and global context modeling. For global context, the model predicted for each image in the ordered group whether it shows the highest digit in the group and whether any digit is unique within the group. For local context, it predicted whether an image shows the highest digit among its two direct neighbors. Because the images themselves do not indicate position, the model had to rely entirely on positional encoding. To quantify the expected limitation of relative encodings for long-range absolute positions such as the middle of the group, we added two further objectives: whether an image is in the middle or at the outside of the group. The model was trained for all learning objectives simultaneously. For synthetic experiments, we used CoM³eT-Tiny, which used a Swin-Tiny vision backbone with ImageNet-1k weights[41]. Its image transformer had 2 layers, 8 attention heads, and a hidden size of 256. All models were trained for 100 epochs. To isolate the effect of context and positional information, we compared three settings: First, we processed each patch token independently, with no context. Second, we enabled the image transformer but disabled positional encoding. This meant that the model could share information across patch tokens, but could not determine their relative order or distance. Third, we used both the image transformer and our symmetric variant of relative ALiBi positional encoding.

We evaluated the image transformer on three representative real-data sparse tasks, chosen to span different context requirements. In pathology, "Cancer Recurrence (Whole Slide)" and "Surgery Outcome (Whole Slide)" did not use positional encoding and represented each image by a large number of patch tokens. In radiology, "Tumor Classification (MRI)" used positional encoding. In each case, we compared CoM³eT against a 2D variant without the image transformer. Without the image transformer (2D processing), the model operated on individual patches, so we averaged the result over all patches.

For dense multidimensional prediction, the pyramid transformer used the output of the image transformer to compute relationships between feature maps across a multidimensional image. It can be applied wherever a feature map is computed, however, it is desirable to learn such complex operations in the shared modules, not in the task-specific modules.

The pyramid transformer took as input a feature map F of size (C, H, W) for S patch tokens and the output V of the image transformer of size d for each patch token, and produced a contextualized feature map of the same size. First, it computed relationships A using only the image transformer's output via self-attention with sigmoid activation instead of softmax. Here, Q and K were computed from V via linear layers, with Q divided by $\sqrt{d}$ resulting in the $S \times S$ relationships represented by the attention matrix A. Second, $F'_{i,j}$ for each spatial location were computed independently based on the feature maps of all patch tokens:

$$F'_{i,j} = \underbrace{\texttt{sigmoid}\left(QK^T\right)}_{A} F_{i,j}$$

The feature map F' was further processed by applying a layer normalization with affine parameters `Norm`, a GELU activation, and a learnable layer scale γ. Finally, the result was returned via a residual connection to the input feature map F:

$$\texttt{Output} = F + \gamma\left(\texttt{GELU}\left(\texttt{Norm}\left(F'\right)\right)\right)$$

We used synthetic segmentation tasks to test local context, global context, and patch token-level prompting. We generated 10,000 training ordered groups of images and 1,000 test groups, each with 3 to 8 images of size 224×224 and up to 3 objects per image. Each image showed geometric shapes, including ellipses, rectangles, and X shapes in different colors. We randomly built the groups once to simulate a fixed-length dataset and distributed them across multiple images. Copying masks did not overwrite non-background original masks. The local-context task required the model to segment ellipses in the image where they appeared and in the direct neighboring images, where the ellipse itself was not visible. The global-context task required the model to segment X-shapes across a group when X-shapes were the most common object class in that group. The prompting task required the model to globally copy rectangle segmentations across images that carried a "copy" prompt, as illustrated in Extended Data Fig. 1 C. The prompts were represented by SONAR[42] text embeddings, and were added to the patch tokens before the image transformer.

We then evaluated the pyramid transformer on three representative real-data tasks chosen to span different dataset sizes and context requirements: "Uterus Segmentation (MRI)", "Vessel Segmentation (CT)", and "Tumor Segmentation (MRI)". For each task, we compared CoM³eT against a 2D variant without the transformer components and against a specialized reference method. The reference was nnU-Net[10], except for "Uterus Segmentation (MRI)", where nnU-Net did not produce useful results and a 3D U-Net[28] served as the reference instead. Confidence intervals were computed by bootstrapping with 10,000 iterations and sampling with replacement.

Benchmarking CoM³eT for medical imaging: UNICORN

In medical imaging, FMs must be applicable to both sparse tasks (for example, regression and classification) and dense prediction tasks (for example, detection and segmentation). UNICORN[14] was a competition for medical FMs that evaluated frozen models across pathology and radiology, covering dense and sparse as well as 2D, multidimensional, vision, and vision-language tasks. Its task set included 3 classification tasks, 1 regression task, 4 detection tasks, 3 segmentation tasks, and 1 vision-language task, spanning 5 radiology and 7 pathology tasks. Overall, 8 tasks were multidimensional (five 3D and three WSI), 3 were 2D, and 1 was vision-language. To directly compare FMs, it used a frozen setting: FM weights remained fixed during evaluation, and lightweight adapter models were trained on top of extracted representations. The benchmark used 48 few-shot labeled training examples for each task. Training and evaluation ran privately on the Grand Challenge platform.

For our submission, we used CoM³eT-UNICORN. It was identical to CoM³eT-Base, but used a smaller number of dimensions for the hyperpixel token because of memory constraints (RAM) on the evaluation platform. Specifically, we set the dimension of the patch tokens to 512 (unchanged) and the dimension of the hyperpixel tokens to 32. We used these settings across all tasks and domains.

2D tasks were processed by applying CoM³eT's vision backbone to the whole image as a single patch token. For each sparse 3D and WSI task (e.g. classification), up to 10,000 image patches were extracted as patch tokens. Because these tasks required a single feature vector per image, but in our method, the number of feature vectors corresponded to the number of image patches, we selected the most attended patch token from the last attention layer of the image transformer as image-level feature vector and concatenated it with the mean representation of all patch tokens. For 3D segmentation, it was possible to apply the usual multidimensional segmentation pipeline of CoM³eT (vision backbone → image transformer → pyramid transformer → decoder), which produced a 3D feature map with the same spatial dimensions as the input image and a feature dimension of 32.

In this paper, we report the performance of CoM³eT on the UNICORN benchmark, comparing it with the strongest competing model picked for each task, and with results aggregated by domain (pathology and radiology) with the second- and third-strongest other models. For more details on the UNICORN benchmark, please refer to the benchmark publication[14]. For confidence intervals (error bars in Fig. 3 A), we averaged the confidence intervals of the individual tasks in the same way as the mean task-specific metrics: rescaled such that the range from random performance to perfect performance was 0 to 1.

Pretraining database

Training followed a multistage approach: first natural pretraining based on large-scale natural image data, then expanded training including both medical and natural data. Natural image datasets provided a vast amount of diverse visual training data, which is beneficial for learning general visual concepts. Medical image datasets contributed general pretraining data[4,43] and a diverse collection of domain-specific knowledge for medical imaging.

To scale the natural-image domain, we used ImageNet21k[22] with its more than 11 million labeled images across roughly 21,000 classes. ImageNet21k is a strong pretraining dataset but hard to incorporate: its label hierarchy usually demands specialized handling, or labels to be dropped entirely, as in Swin V2's self-supervised pretraining[38]. CoM³eT instead used all images and labels, addressing label ambiguity by passing groups of four examples through the image transformer and predicting the class shared by the group. We also included ImageNet-1k[21] for its higher label quality and established use in pretraining, using its 1,281,167 training images across 1,000 classes restricted to leaf nodes in the ImageNet hierarchy, such as "German shepherd" rather than broader classes such as "mammal". Beyond classification, the database also included natural images with tasks for vision-language alignment, segmentation, and object detection from the COCO database[23], contributing 118,287 images with 591,753 captions for vision-language alignment and more than 200,000 images annotated for the 12 supercategories and 80 object categories. For the vision-language task, negative examples for the discriminator were sampled from other images.

For digital pathology, the database covered patch-level tissue characterization, tumor detection and grading, mitosis detection, and dense tissue segmentation across colorectal, breast, and prostate pathology. CoM³eT was trained with 100,000 H&E tiles from colorectal cancer with a classification task over nine tissue classes such as adipose, lymphocytes and normal mucosa[44], 682 gastric image regions from 155 patients for patch-based segmentation of gastric mucosa and intestine tissue[45], 213 colorectal adenocarcinoma regions of interest with gland annotations for gland segmentation[46], 4,981 colon regions of interest with about half a million segmented nuclei[47], and 20 whole-slide images with 10 tissue classes for tissue segmentation[48]. Regarding breast pathology, the database included 405 regions of interest with 9,501 mitotic figure annotations for both detection and segmentation[49], 151 whole-slide images with 18 tissue classes such as tumor and stroma[50], and 370 whole-slide images with seven classes such as invasive tumor, in-situ tumor, and inflamed stroma[51]. For ovarian pathology, tumor subtype patch- and image-level annotations were included[52]. In prostate pathology, it included more than 10,000 whole-slide images for tumor grading and tissue-type segmentation[53], around 2,000 patients for biochemical recurrence prediction from public[54,55] and private sources, epithelium segmentation[56], basal-cell segmentation (private), prostate histopathology classification with Gleason grading[57,58], and about 200,000 additional tiles for tumor detection[59,60]. Other data included captions mined from pathology publications, included as a vision-language task[4], and tumor proportion score prediction for IHC stained lung tissue[61].

Because CoM³eT processes stain types beyond H&E, foreground extraction through simple thresholding was not possible. However, sampling within tissue foreground tiles during pretraining was necessary for efficiency. Furthermore, we wanted to exclude pen annotations and regions with heavy artifacts. We used the partial fine-tuning setting with earlier versions of CoM³eT, and iteratively fine-tuned a multitask model for foreground and artifact detection. In each iteration, we annotated one image, preferably by correcting a model mistake while leaving most of the image unlabeled, then partially fine-tuned the model for a few seconds using 25 update steps with all annotated data collected so far, and finally predicted on another random whole-slide image to identify where the next annotation was needed. The final foreground model was trained with 147 sparsely annotated zoomed out images and used for sampling of foreground regions in all histology datasets.

When loading a tomographic image, training used 3D patches with a probability of 80%, and otherwise full volumes. Patches had a randomly chosen depth between 1 and 32 slices and were sampled such that the corresponding volume contained at least one foreground voxel when possible. The patch edge length depended on the task and was most commonly between 128 and 384 pixels. To keep data loading from becoming a bottleneck in the main training process, we used group caching with an adaptive cache rate. We first loaded a group of image patches, applied augmentations that required the whole group, including random 3D rotations, resampling, cropping, scaling, and intensity augmentations, and then cached the result. We then loaded a subgroup from the cache and applied local augmentations, including rotations in 90 degree steps, additional intensity augmentations, and distortions. To pretrain the model to extract general features in radiology, RadImageNet[43] was used. This dataset contributes over one million images of various anatomical structures and pathologies.

For multidimensional segmentation, the pretraining database included TotalSegmentator[62], a collection of 3D CT segmentation masks for 24 organs, 26 vertebrae, 18 cardiac structures, 23 muscles, and 26 ribs with five segmentation tasks (one per category), and the Medical Segmentation Decathlon[63] with 10 3D segmentation tasks. Further, it included anatomic structure segmentations for prostate[64], abdominal organs in CT[65], spinal structures in MRI[66].

For lesion-related pretraining, we used public data with 32,120 lesions using rectangle annotations with an object detection task in CT[67], 7,095 regions of interest around lesions with a segmentation task for CT[68], segmentation masks of 1,507 patients for breast MRI[69] and 369 patients for brain tumor in MRI[70]. We also included private full-body CT lesion segmentation from a melanoma cohort[71,72], comprising 4,727 lesions from 262 patients with stage IV malignant melanoma, including lung (2,006), liver (753), soft tissue or skin (1,155), lymph node (379), skeletal (123), spleen (99), and other lesions (212). Lesion malignancy classification data were included for breast MRI[73], and CT[74,75].

Further patch-level data broadened the database. For chest radiography, we included anatomical segmentation[76], pneumothorax segmentation[77], and identification of common thoracic findings[78,79]. Beyond radiology, the database covered cell-level classification of bone marrow cytology in hematologic malignancies[80] and polyp segmentation in gastrointestinal endoscopy[81].

Training method

The training method was built on UMedPT[13] with key changes. Because pretraining alternates between many tasks, keeping every task head and its gradients in memory would make the memory footprint grow with the number of tasks. Instead, each step processed one task at a time: after a task's turn, we applied the update to its head and immediately discarded the task-specific gradients before moving to the next task. As a result, memory requirements stayed constant enabling a larger batch size and more tasks with the same hardware.

Tasks with structured labels such as classification used a single linear layer, and segmentation tasks used a single convolutional layer. Besides segmentation and classification, CoM³eT supported vision-language pretraining. For this purpose, we used SONAR[42] as text model, which can both compute sentence embeddings and generate text from embeddings. Captions were augmented using the Qwen-3 chat model[82]. The training objective was to predict the caption from the image and discriminate correct captions from incorrect ones, inspired by CoCa[83]. Like CoCa, our approach was trained for both captioning and caption discrimination. Unlike CoCa, which jointly trains the text decoder and vision encoder, we aligned with a frozen pretrained unimodal text decoder to free GPU memory and avoid catastrophic forgetting. Our primary objective was pretraining and not captioning performance, so we used a high loss weight for caption discrimination (80%) and only a low weight for caption generation (20%).

For CoM³eT's vision backbone, we found that fp16 produced NaN losses with the Swin Transformer, which led to the poor performance shown in Extended Data Fig. 1, and therefore used bfloat16. Together, mixed precision and gradient checkpointing enabled training with 2× larger batch sizes, which was important for processing large multidimensional images with CoM³eT-Large. CoM³eT-Large and Base variants used bfloat16. CoM³eT-Tiny used full precision (fp32).

For multi-node training, we used the ZeRO optimizer[84], which partitions optimizer state across processes so that no single device holds a full copy. This reduces per-device memory relative to standard Adam, whose optimizer states require roughly twice the model size. Each node processed a subset of tasks, accumulated local gradients, and averaged them across nodes via PyTorch's DistributedDataParallel using NCCL for inter-GPU communication. Auxiliary task heads were kept fully local, while shared model components were managed by DDP. Training ran on 10 A100 80GB GPUs across five machines, each with 334 GB RAM and 24×2.20 GHz Intel Xeon Gold 5320 CPUs.

To train with many large datasets, several technical advancements were implemented. Gradient checkpointing was used around each block of the vision backbone. Two variants of gradient checkpointing were tested to reduce memory requirements during training: one checkpoint around the entire vision backbone, which has the advantage of being simple to add to any architecture, and one checkpoint around each encoder segment as done in[38], which is more memory efficient but requires more implementation effort when using other types of encoders. Both variants were tested with synthetic data for classification, segmentation, group classification, and group segmentation tasks.

Partial fine-tuning of CoM³eT

We explored partial fine-tuning as a parameter-efficient fine-tuning strategy for CoM³eT. Specifically, we froze the vision backbone and trained the image transformer, pyramid transformer, and decoder modules. For patch-level tasks, the image transformer was equivalent to a pretrained MLP because the group size was one, and classification tasks did not use the decoder. The resulting patch tokens were cached once and reused during training.

We compared full and partial fine-tuning. To do so, we selected representative downstream tasks that covered the main task types. "Uterus Segmentation (MRI)" was chosen for 3D segmentation, and "Surgery Outcome (Whole Slide)" and "Cancer Recurrence (Whole Slide)" were chosen for gigapixel image analysis. For 2D tasks, "Pneumonia (X-ray)" represented classification and "Tumor Segmentation (Ultrasound)" and "PCLS Segmentation (Microscopy)" represented segmentation. "Uterus Segmentation (MRI)", "Tumor Segmentation (Ultrasound)" and "PCLS Segmentation (Microscopy)" tested for strong generalization because no similar data were included in pretraining.

Each update step sampled eight patients for multidimensional tasks and 64 patients for patch-based tasks. All trainings used 15 loops with 100 update steps each. Up to 2,500 patch tokens per patient were cached. Because full fine-tuning is memory-intensive and we wanted identical settings across both strategies, we subsampled 50 to 100 tokens per patient per update step whenever more were available. Unless stated otherwise, we used cross-validation with five splits of 70% training and 30% validation data, repeated three times, yielding 15 samples per experimental condition.

Federated fine-tuning of CoM³eT

We implemented federated learning with Flower[85] to use CoM³eT with federated fine-tuning for a single downstream application. The FM in this experiment was CoM³eT-Base without biochemical recurrence-specific pretraining data. Starting from this pretrained checkpoint, we performed federated fine-tuning to obtain CoM³eT-FL. As in the partial fine-tuning experiment, we fine-tuned the image transformer while keeping the vision backbone frozen, and each site kept its patient data local and shared only updates for the trainable parameters. We used a standard server-based optimization scheme similar to Federated Averaging[6]: Each site trained locally for a short interval, sent parameter updates for aggregation, and received the updated global model from the server. Because the transferable part of the model was small, we synchronized after every 4 local update steps rather than after full local epochs.

We ran this setup in a real multi-site scenario with three sites: two German university hospitals and a research institute that pooled relevant public data. The sites used consumer-grade GPUs and weak internet connections under real-world hospital IT constraints. Communication between nodes used gRPC over HTTP/2. We used a containerized AI environment that isolated both patch token caching and federated fine-tuning from the host system and allowed each participant to define separate read and write storage resources to meet data-protection requirements. For patch-token caching, Charité used four Tesla T4 GPUs, and UKFFM used four RTX A5000 GPUs. Federated fine-tuning then ran on a single Tesla T4 GPU at Charité and a single RTX A5000 GPU at UKFFM.

Statistics

To compare full and partial fine-tuning, we used equivalence testing with two one-sided tests (TOST) between task and training-split pairs, ruling out that one strategy was meaningfully worse than the other by a predefined margin of 0.05 for the main performance metric of each task (for example, AUC for classification and C-index for time-to-event prediction). Each side was a Wilcoxon signed-rank test on the paired differences. We considered two strategies equivalent if both p_{lower} and p_{upper} were below 0.05, and used the same test to compare federated with centralized fine-tuning. To assess whether attention topology augmentation outperformed full attention when fine-tuning, we used a one-sided paired t-test, pairing observations by task and cross-validation split across the two whole-slide tasks ($n = 10$ pairs), and verified the normality assumption of this test with a Shapiro-Wilk test on the paired differences.

Test data

Cancer Recurrence (Whole Slide) "Cancer Recurrence (Whole Slide)" is a time-to-event prediction task for biochemical recurrence after prostatectomy. The quality of learned representations was evaluated with PRAD multi-center data[24] split into five centers. Four centers (N=97, N=86, N=85, N=71) were kept as is, while the remaining centers, which often contained only one sample each, were pooled into one center (220 samples from 27 sites). This task was included to enable the comparison of models for their application on datasets with many patch tokens per patient.

Surgery Outcome (Whole Slide) "Surgery Outcome (Whole Slide)" is a whole-slide image classification task that predicts whether biochemical recurrence occurs within one year after prostatectomy based on the CHIMERA challenge data[25]. It contains data from 94 patients with 190 images. This task complemented "Cancer Recurrence (Whole Slide)" to compare models for whole-slide image analysis with many patch tokens per patient.

Uterus Segmentation (MRI) "Uterus Segmentation (MRI)" is a volumetric segmentation task for three uterine structures: myometrium, junctional zone, and endometrium. All structures were segmented up to the cervical junction by a radiologist. The dataset is part of the RACOON FADEN project on early detection of adenomyosis and comprises 13 female subjects (27 ± 5.44 years, BMI 21.8 ± 2.96) collected across three German university hospitals. T2-weighted short-axis uterine MRI were acquired with motion-insensitive, multi-shot TSE BLADE sequences, yielding an anisotropic voxel spacing of $0.643 \times 0.643 \times 3.0$ mm. Methods were compared over five repetitions on the same test patients used in the dataset's publication[28], and compared with a 3D U-Net as the baseline method. nnU-Net was also evaluated but did not produce useful results. This task was included to compare models for 3D segmentation in a small-data setting, where FMs are expected to be particularly useful.

PCLS Segmentation (Microscopy) "PCLS Segmentation (Microscopy)" assessed the segmentation of precision-cut lung slice (PCLS)[86] tissue samples in brightfield microscopy images. PCLS are an ex vivo organotypic model prepared by slicing agarose-inflated lung tissue with a vibratome, preserving the three-dimensional cellular architecture, resident cell populations, and cell–matrix relationships of the native lung. Recent advances in culture conditions have enabled long-term (multi-week) cultivation of viable PCLS, opening the door to longitudinal monitoring of tissue behavior over extended time periods. This, in turn, permits the analysis of fibrotic tissue remodeling and treatment response over time[87,88]. For such applications, accurate segmentation of the tissue piece itself is the first processing step, as it enables analyses of relative tissue growth or shrinkage over time (a potential marker of fibrosis progression) as well as masking of subsequent tissue analyses. Tissue segmentation in these microscopy images is non-trivial due to various challenges such as recording artifacts (e.g. poorly focused recordings), ambiguous tissue boundaries, shadows, and formation of air bubbles. The PCLS microscopy recordings were obtained using a Zeiss Axio Observer Z1 brightfield microscope by a trained laboratory technician. For tissue segmentation, recordings were downscaled to 512×512 and converted to JPGs. Tissue and background partial annotations were created using LabelStudio for 86 sample recordings. This task was used to assess the ability of partial fine-tuning to generalize to a new modality.

Tumor Segmentation (Ultrasound) "Tumor Segmentation (Ultrasound)" is a 2D lesion segmentation task consisting of 256 B-mode ultrasound scans from 256 patients[89]. Of these scans, 154 show benign tumors, 98 show malignant tumors, and four show normal tissue. Each scan was manually segmented by five experienced radiologists. The dataset is skewed towards smaller tumors, supporting the development of methods for earlier detection. This task tested generalization because no ultrasound-specific pretraining data was included.

Pneumonia (X-ray) "Pneumonia (X-ray)"[90] is a chest X-ray classification task for pneumonia diagnosis containing 5856 pediatric images, each labeled as either normal or pneumonia. It was included as the 2D classification task in the partial fine-tuning experiments and enabled this study to test weak generalization, because while no pediatric images were included, chest X-ray data were already included in pretraining of CoM³eT.

Vessel Segmentation (CT) "Vessel Segmentation (CT)" is a binary segmentation task for the aorta and its branches in 3D CTA images, selected as a 3D setting expected to depend strongly on local continuity across neighboring slices. Training data[91] consisted of 100 CTA images with axial dimensions ranging from 389×389 to 516×516 pixels (average 450×450), isotropic voxel resolution of 1 mm, and 578 to 801 axial slices per image (average 695). Test data[92] comprised 56 CTA scans from mostly healthy aortas, covering the aortic arch and its branches and the abdominal aortas with the iliac arteries, with expert binary segmentation masks. The collection included one case with an abdominal aortic aneurysm and five cases with aortic dissections. nnU-Net[10] was used as the baseline method. This task was included to compare models for a task that requires volumetric continuity.

Tumor Segmentation (MRI) "Tumor Segmentation (MRI)" is a 3D segmentation task for breast lesions in dynamic contrast-enhanced MRI, covering malignant, benign, and lesions of unknown histology with a mean lesion volume of 6.1 ± 20.3 ml. The MRI protocols included acquisitions with and without fat suppression on 1.5T and 3T scanners from GE, Siemens Healthineers, and Philips Healthcare, with varying voxel sizes and fields of view. nnU-Net[10] was used as the baseline method. This task was included to compare the 3D segmentation capabilities of models for which multi-center robustness and data efficiency are irrelevant due to the inclusion of a large dataset from multiple hospitals.

Tumor Classification (MRI) The "Tumor Classification (MRI)" task distinguished between benign and malignant breast lesions in dynamic contrast-enhanced MRIs based on data from a private Siemens Healthineers collection. The collection spanned more than ten clinical institutions, including 1,330 imaging studies from 1,299 patients and 1,796 lesions with an average volume of 4.8 ± 14.8 ml. The malignant class included lesions that were histologically confirmed as malignant or judged as obviously malignant by a radiologist (BI-RADS 5). The benign class included lesions that were histologically confirmed as benign or obviously benign (BI-RADS 2) and therefore not biopsied. A video Swin Transformer[27] trained on small regions of interest around the lesions was selected as the baseline method due to its high performance, multidimensional pretraining based on the Kinetics video dataset, and open-weight availability[41]. This large, multicenter dataset was used to create a task that favors specialized, single-task baselines for 3D classification, testing whether CoM³eT's unified architecture would remain competitive.

Data availability

The following test datasets are publicly available: "Cancer Recurrence (Whole Slide)"[24], "Surgery Outcome (Whole Slide)"[25], "Vessel Segmentation (CT)"[91,92], "Tumor Segmentation (Ultrasound)"[89], and "Pneumonia (X-ray)"[90]. The following test datasets can be obtained from the corresponding author upon reasonable request: "Uterus Segmentation (MRI)" with additional permission from the RACOON consortium, "Tumor Segmentation (MRI)" and "Tumor Classification (MRI)" with additional permission from Siemens Healthineers, and "PCLS Segmentation (Microscopy)" with additional permission from Fraunhofer ITEM.

Code availability

The code is publicly available at https://github.com/FraunhoferMEVIS/MedicalMultitaskModeling.

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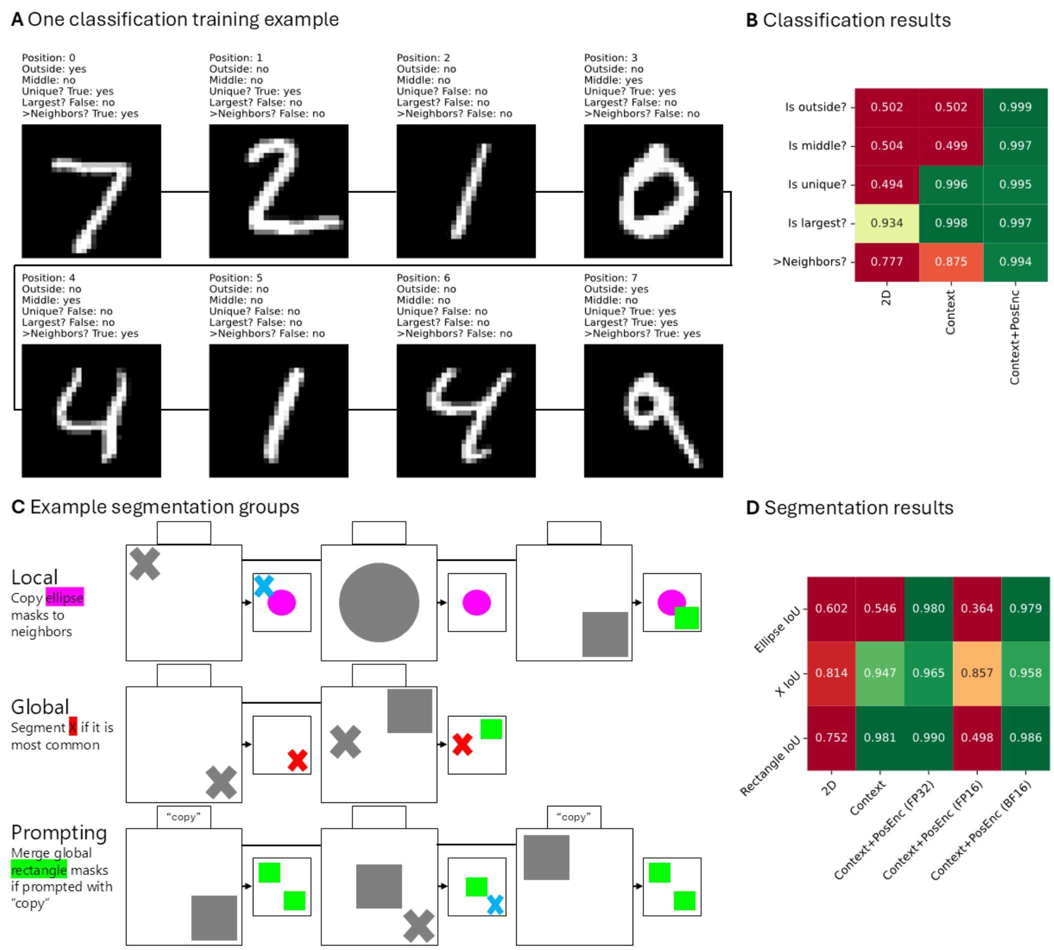

Extended Data Fig. 1: Synthetic data. A, A single example ordered group of 8 images. **B,** AUC of a 2D model, a model with the image transformer but without positional encoding (Context), and the full model with the image transformer and positional encoding as in CoM³eT (Context+PosEnc). **C,** Three example groups. Each image shows the input image (large square), an optional language prompt (small box), and the corresponding segmentation mask (small square). The images are shown in gray so that color indicates the ground-truth segmentation. In this task, each image contains 0 to 3 objects with varying color, size, and location, and each group contains 3 to 8 images. **D,** Performance of a 2D model (2D), a model with the image transformer and the pyramid transformer but without positional encoding (Context), and the full model with additional positional encoding as in CoM³eT (Context+PosEnc). For the full model, we show results in full precision, half precision, and bfloat16. Synthetic tasks were used to verify that CoM³eT was able to learn sparse and dense multidimensional tasks that required both local and global context, and that ALiBi positional encoding enabled the model to use position information that is not visible in the images themselves.

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Author contributions

J.R.S. prepared the test data, together with K.G., H.v.B. and R.G. (Tumor Segmentation (MRI) and Tumor Classification (MRI)); C.T., M.M., L.S. and S.A. (Uterus Segmentation (MRI)); K.H. (Vessel Segmentation (CT)); and E.P., A.P. and N.A. (PCLS Segmentation (Microscopy)). J.R.S., K.G., K.H. and E.P. analysed the results. J.R.S., T.N., T.B., I.Do., R.M., N.F., P.W. and N.Z. ran the federated-learning experiments. J.R.S. and H.M. built the pretraining infrastructure, and J.R.S. and N.W. developed the software for data curation. J.R.S., T.N., L.O.S., A.G., J.H.M., F.P., I.Da., T.R.K. and S.E. prepared the pretraining data. P.W., T.R.K. and S.E. contributed as clinical advisors for pathology, and F.K., F.P. and I.Da. as clinical advisors for radiology. J.R.S. wrote the paper, with feedback from all authors. F.K. and J.L. contributed equally and coordinated the study.

Competing interests

The applicant 'Fraunhofer-Gesellschaft zur Förderung der angewandten Forschung eingetragener Verein' has a patent pending related to the training algorithm and neural architecture components presented in this article (patent application no. EP23209015.9; names of inventors, J.R.S., T.N., J.L., F.K.).

Supplementary information

Partial fine-tuning of CoM³eT performs competitively in predicting biochemical recurrence in prostate cancer

We submitted an early version of CoM³eT to the LEOPARD study[54] using partial fine-tuning. This model had been pretrained on 42 tasks including prostate-cancer-related tasks such as ISUP grading and tumor detection in whole-slide images but did not include biochemical-recurrence-specific data. We then partially fine-tuned it in a multitask setting that included the LEOPARD training data and the PRAD cohort[24], each with its own task head. The study evaluated four cohorts of varying size, each contributed by a single hospital, and compared other FMs and specialized algorithms in this external setting.

CoM³eT achieved the best result on a single private cohort with a C-index of 0.771 (95% CI: 0.727 to 0.813). The second-best result was 0.714 (95% CI: 0.669 to 0.761) based on the pathology FM CTransPath[93]. Across all four cohorts, with $N = 100$ test patients from the training cohort and $N = 723$, $N = 427$, and $N = 332$ from the external cohorts, CoM³eT reached a C-index of 0.707, tied with tumor grading as currently used in clinical practice. When weighted by the number of test patients in each cohort, CoM³eT achieved a C-index of 0.720, ahead of the next best results of other participants with 0.704 and 0.699.